

Echelon Forms

Geometric Algorithms Lecture 2

Practice Problem

None today...

3 planes in $\mathbb{R}^3$, none parallel, but no solution
(no single point where all intersect)

Outline

- » Introduce echelon forms as matrices which "look like" solutions
- » Learn to "read off" a solution from an echelon form

Keywords

leading entries

echelon form

(row-)reduced echelon form (RREF)

pivot positions

pivot columns

free variables

basic variables

general form solutions

forward elimination

back substitution

Recap

Recall: Linear Systems (General-form)

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

Recall: Linear Systems (General-form)

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

Does a system have a solution?

How many solutions are there?

What are its solutions?

Recall: Matrix Representations

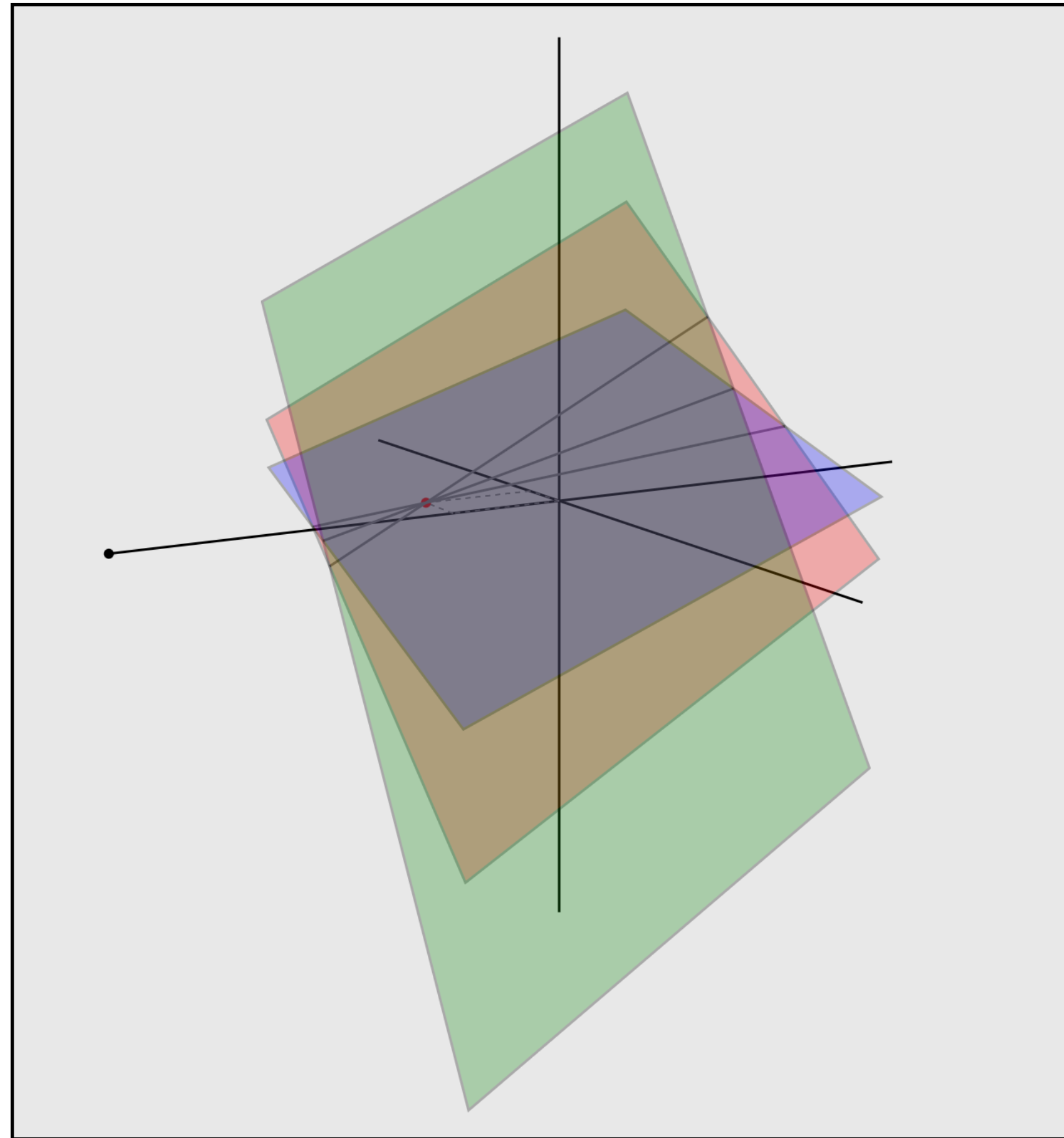
$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

Recall: Matrix Representations

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

augmented matrix

Recall: Linear Systems, Geometrically



Recall: Number of Solutions

zero the system is inconsistent

one the system has a unique solution

many the system has infinity solutions

Recall: Number of Solutions

zero the system is inconsistent

one the system has a unique solution

many the system has infinity solutions

These are the **only** options

where we left off...

Solving Systems with Two Variables

$$2x + 3y = -6$$

$$4x - 5y = 10$$

Another Approach

Elimination

Eliminate x from the EQ2 and solve for y

Eliminate y from EQ1 and solve for x

Back-Substitution

Solving Systems with Two Variables

$$2x + 3y = -6 \quad \text{EQ}_1$$

$$4x - 5y = 10 \quad \text{EQ}_2$$

$$\begin{array}{r} -4x \\ -6y \\ +12 \end{array}$$



$$2x + 3y = -6$$

$$-11y = 22$$

$$\begin{array}{r} 3y \\ -6 \end{array}$$

$$\begin{array}{r} 2x + 3y = -6 \\ y = -2 \end{array}$$

$$\text{EQ}_2 - 2\text{EQ}_1$$

$$\text{EQ}_2 (-11)$$

$$\text{EQ}_1 - 3\text{EQ}_2$$

Forward elimination

still a linear system

$$2x + 3y = -6$$

$$4x - 5y = 10$$

+

$$6x - 2y = 4$$

$$2x = 0$$

$$y = -2$$

$$\text{EQ}_2 / 2$$



$$x = 0$$

$$y = -2$$

Solving Systems as Matrices

How does this look with matrices?

Observation. Each intermediate step of elimination and back-substitution gives us a new linear system with the same solutions

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Observation. Each intermediate step of elimination and back-substitution gives us a new linear system with the same solutions

Can we represent these intermediate steps as operations on matrices?

Let's look back at this...

$$\begin{matrix} R_1 \\ R_2 \end{matrix} \begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \\ -4 & -6 & +12 \end{bmatrix}$$

$$R_2 \leftarrow R_2 - 2R_1$$

$$2x + 3y = -6$$

$$-11y = 22$$

$$R_2 \leftarrow R_2 / -11$$

$$R_1 \leftarrow R_1 - 3R_2$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 0 & -11 & 22 \\ 0 & -3 & +6 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 0 & 1 & -2 \end{bmatrix}$$

$$2x + 3y = -6$$

$$4x - 5y = 10$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

$$R_1 \leftarrow R_1 / 2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

$$x = 0$$

$$y = -2$$

Elementary Row Operations

scaling	multiply a row by a NONZERO number
replacement	add a multiple of one row to another
interchange	switch two rows

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scaling	multiply a row by a NONZERO number
replacement	add a multiple of one row to another
interchange	switch two rows

These operations don't change the solutions

Scaling

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow{R_3 \leftarrow 3R_3} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 21 & 24 & 27 \end{bmatrix}$$

Scaling

$$\begin{array}{l} 2x + 3y = -6 \\ 4x - 5y = 10 \end{array}$$

$$R_1 \leftarrow 2R_1$$



$$\begin{array}{l} 4x + 6y = -12 \\ 4x - 5y = 10 \end{array}$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 6 & -12 \\ 4 & -5 & 10 \end{bmatrix}$$

Replacement

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 + 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 6 & 9 & 12 \\ 7 & 8 & 9 \end{bmatrix}$$

Handwritten red annotations above the first matrix: $+2$ above 4, $+4$ above 5, and $+6$ above 6.

Replacement

$$\begin{array}{l} 2x + 3y = -6 \\ 4x - 5y = 10 \end{array}$$

$$R_2 \leftarrow R_2 + R_1$$



$$\begin{array}{l} 2x + 3y = -6 \\ 6x - 2y = 4 \end{array}$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 6 & -2 & 4 \end{bmatrix}$$

Interchange

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 7 & 8 & 9 \\ 4 & 5 & 6 \\ 1 & 2 & 3 \end{bmatrix}$$

Interchange

$$\begin{array}{l} 2x + 3y = -6 \\ 4x - 5y = 10 \end{array}$$

$$R_1 \leftrightarrow R_2$$



$$\begin{array}{l} 4x - 5y = 10 \\ 2x + 3y = -6 \end{array}$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 4 & -5 & 10 \\ 2 & 3 & -6 \end{bmatrix}$$

Example: Row Reductions

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 2 & 3 & -6 \\ 0 & -11 & 22 \end{bmatrix}$$

Example: Row Reductions

$$\begin{array}{ccc} & R_2 \leftarrow R_2 - 2R_1 & \\ \begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix} & \longrightarrow & \begin{bmatrix} 2 & 3 & -6 \\ 0 & -11 & 22 \end{bmatrix} \\ & R_2 \leftarrow R_2 / (-11) & \\ & \longrightarrow & \begin{bmatrix} 2 & 3 & -6 \\ 0 & 1 & -2 \end{bmatrix} \end{array}$$

Example: Row Reductions

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

$$R_2 \leftarrow R_2 - 2R_1$$



$$\begin{bmatrix} 2 & 3 & -6 \\ 0 & -11 & 22 \end{bmatrix}$$

$$R_2 \leftarrow R_2 / (-11)$$



$$\begin{bmatrix} 2 & 3 & -6 \\ 0 & 1 & -2 \end{bmatrix}$$

$$R_1 \leftarrow R_1 - 3R_2$$



$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

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$$\begin{bmatrix} 2 & 3 & -6 \\ 0 & 1 & -2 \end{bmatrix}$$

$$R_1 \leftarrow R_1 - 3R_2$$



$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

$$R_1 \leftarrow R_1 / 2$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

Example: Row Reductions

$$R_2 \leftarrow R_2 - 2R_1$$

$$R_2 \leftarrow R_2 / (-11)$$

$$R_1 \leftarrow R_1 - 3R_2$$

$$R_1 \leftarrow R_1 / 2$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

Example: Row Reductions

$$R_2 \leftarrow R_2 - 2R_1$$

$$R_2 \leftarrow R_2 / (-11)$$

elimination

$$R_1 \leftarrow R_1 - 3R_2$$

$$R_1 \leftarrow R_1 / 2$$

back-substitution

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

Row Equivalence

Df. Two matrices are *row equivalent* if one can be transformed into the other by a sequence of row operations

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

Row Equivalence

$$A \sim B$$

Df. Two matrices are *row equivalent* if one can be transformed into the other by a sequence of row operations

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

We can compute solutions by sequence of row operations

IMPORTANT

Row equivalent augmented
matrices represent linear system
with the same solution set

Example

Exercise

$$x - 2y + z = 5$$

$$2y - 8z = -4$$

$$6x + 5y + 9z = -4$$

Q: When are we done reducing?
What's the "target" matrix?

A: when we are able to
"read off" a solution

Motivating Questions

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What matrices "represent solutions"? (solutions are easy to "read off"?)

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Motivating Questions

echelon forms

What matrices "represent solutions"? (solutions are easy to "read off"?)

How does the number of solutions affect the shape of these matrices?

How do we use row operations to get to those matrices?

Unique Solution Case

Unique Solution Case

$$2x - 3y + 5z = 11$$

$$\begin{bmatrix} 2 & -3 & 5 & 11 \\ 2 & -1 & 13 & 39 \\ 1 & -1 & 5 & 14 \end{bmatrix}$$

$\sim$

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

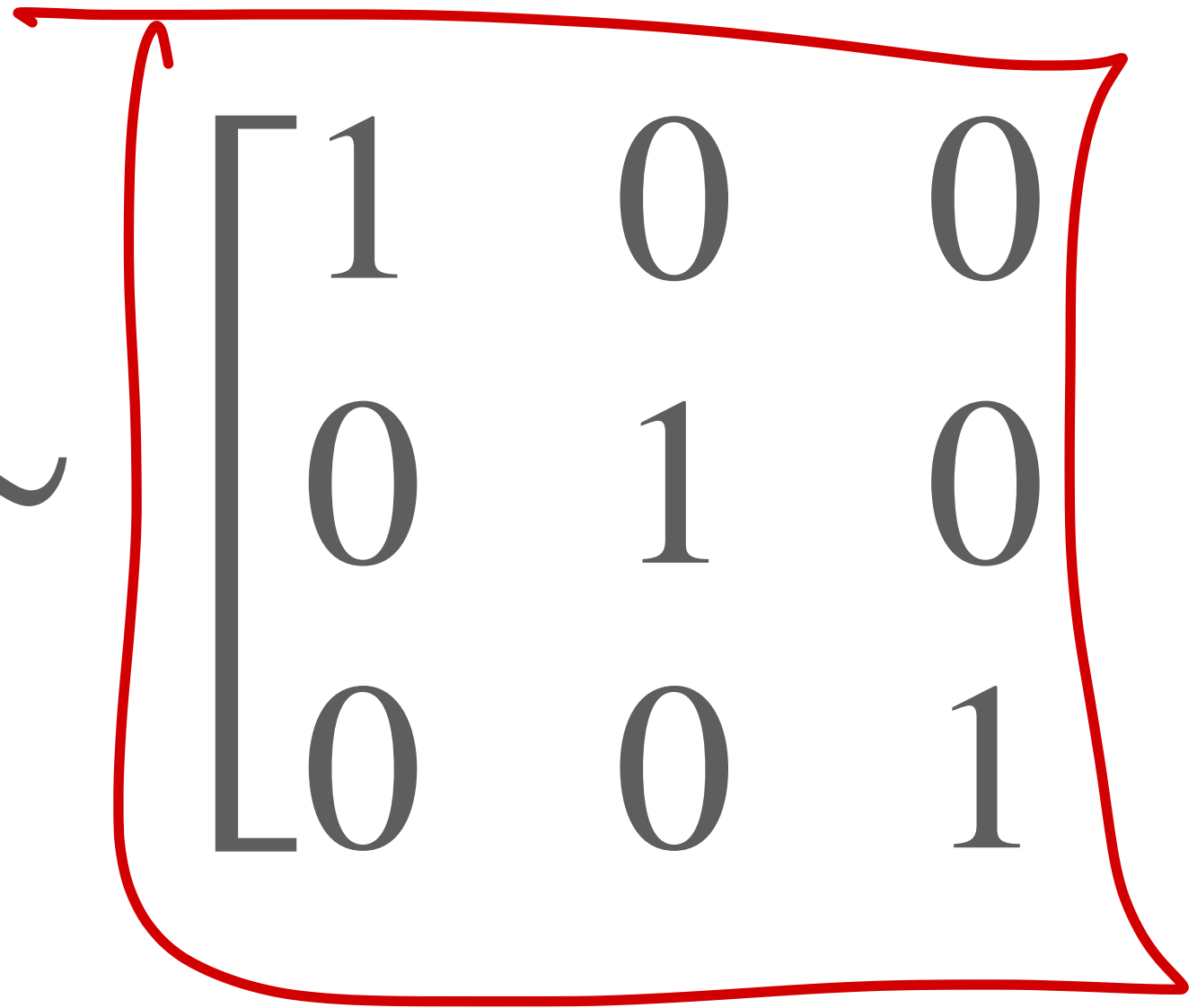
$$x = 1$$

$$y = 1$$

$$z = 2$$

$$z = 3$$

Unique Solution Case

$$\begin{bmatrix} 2 & -3 & 5 & 11 \\ 2 & -1 & 13 & 39 \\ 1 & -1 & 5 & 14 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$


$$x = 1$$

$$y = 2$$

$$z = 3$$

The Identity Matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The Identity Matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

1s along the diagonal

0s elsewhere

Unique Solution Case

coefficient matrix

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

a system of linear equations whose **coefficient matrix** is the identity matrix represents a unique solution

No Solution Case

No Solution Case

$$\begin{array}{l} x+y+z = 1 \\ x+y+z = 2 \end{array} \quad \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$0 = 1$$

No Solution Case

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

two parallel
planes

No Solution Case

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

two parallel planes

row representing $0 = 1$
 $0x + 0y + 0z = 1$

No Solution Case

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

row representing $0 = 1$

a system with no solutions can be reduced to a
matrix with the row

$$0 \ 0 \ \dots \ 0 \ 1$$

Infinite Solution Case

Infinite Solution Case

$$\begin{bmatrix} 2 & 4 & 2 & 14 \\ 1 & 7 & 1 & 12 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$2x + 4y + 2z = 14$$

$$x + 7y + z = 12$$

$$(\pi, 1, 2 - \pi)$$

$$x + z = 2$$

$$y = 1$$

$$(0, 1, 2)$$

$$(-2, 1, 4)$$

Infinite Solution Case

$$\begin{bmatrix} 2 & 4 & 2 & 14 \\ 1 & 7 & 1 & 12 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$x_1 + x_3 = 2$$

$$x_2 = 1$$

Infinite Solution Case

$$\begin{bmatrix} 2 & 4 & 2 & 14 \\ 1 & 7 & 1 & 12 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$x_1 + x_3 = 2$$

$$x_2 = 1$$

a system with infinity solutions can be
reduced to a system which leaves a
variable unrestricted

Infinite Solution Case

$$\begin{aligned}x_1 + x_3 &= 2 \\ x_2 &= 1\end{aligned}$$

it doesn't matter
what x_3 is if we
want to satisfy
this system of
equations

Infinite Solution Case

$$x_1 + x_3 = 2$$

$$x_2 = 1$$

it doesn't matter
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want to satisfy
this system of
equations

$$x_1 = 2$$

$$x_2 = 1$$

$$x_3 = 0$$

Infinite Solution Case

$$\begin{aligned}x_1 + x_3 &= 2 \\x_2 &= 1\end{aligned}$$

it doesn't matter
what x_3 is if we
want to satisfy
this system of
equations

$$x_1 = 1.5$$

$$x_2 = 1$$

$$x_3 = 0.5$$

Infinite Solution Case

$$x_1 + x_3 = 2$$

$$x_2 = 1$$

it doesn't matter
what x_3 is if we
want to satisfy
this system of
equations

$$x_1 = 20$$

$$x_2 = 1$$

$$x_3 = -18$$

Infinite Solution Case

$$x_1 + x_3 = 2$$

$$x_2 = 1$$

it doesn't matter
what x_3 is if we
want to satisfy
this system of
equations

x_1 is free

$$x_2 = 1$$

$$x_3 = 2 - x_1$$

$$x_1 = 2 - x_3$$

$$x_2 = 1$$

x_3 is free

Infinite Solution Case

$$\begin{aligned}x_1 + x_3 &= 2 \\x_2 &= 1\end{aligned}$$

it doesn't matter
what x_3 is if we
want to satisfy
this system of
equations

$$\begin{aligned}x_1 &= 2 - x_3 \\x_2 &= 1 \\x_3 &\text{ is free}\end{aligned}$$

general form

In Sum

- none** reduces to a system with the equation $0 = 1$
- one** reduces to a system whose coefficient matrix is the identity matrix
- infinity** reduces to a system which leaves a variable unrestricted

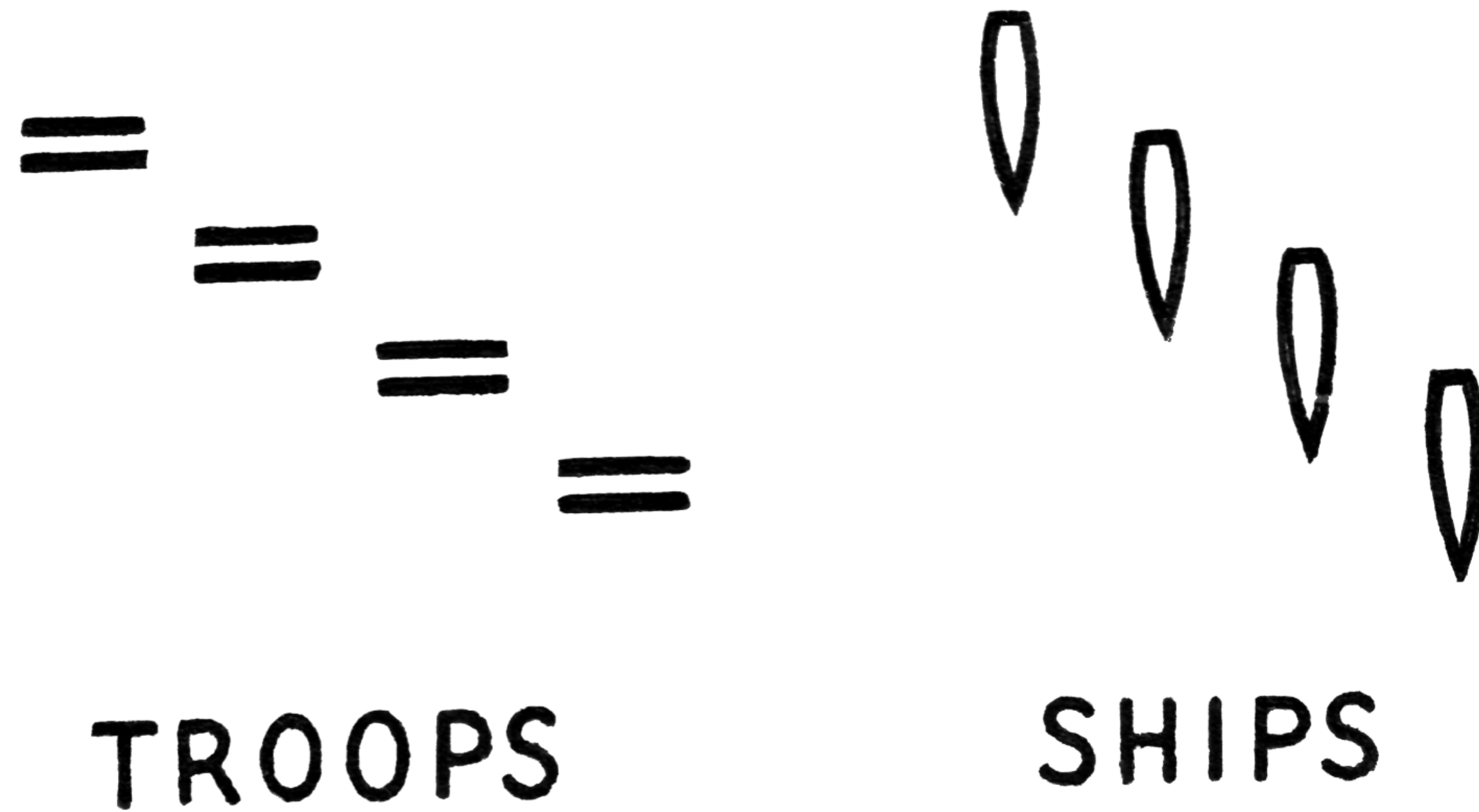
In Sum

- `none` reduces to a system with the equation $0 = 1$
- `one` reduces to a system whose coefficient matrix is the identity matrix
- `infinity` reduces to a system which leaves a variable unrestricted

Ideally, we want one *form* that handles all three cases

Echelon Form

The Picture (and a bit of history)



Echelon Form (Pictorially)

$$\begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\blacksquare$ = nonzero, $*$ = anything

Leading Entries

Df. the *leading entry* of a row is the first nonzero value

$$\begin{bmatrix} 0 & \boxed{1} & 2 \\ 0 & 0 & \boxed{5} \\ \boxed{1} & 2 & 3 \end{bmatrix}$$

Echelon Form

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Df. A matrix is in *echelon form* if

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1. The leading entry of each row appears to the right of the leading entry above it

Echelon Form

Df. A matrix is in *echelon form* if

1. The leading entry of each row appears to the right of the leading entry above it
2. Every all-zeros row appears below any non-zero rows

Echelon Form (Pictorially)

$$\begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\blacksquare$ = nonzero, $*$ = anything

Echelon Form (Pictorially)

next leading entry
to the right

$$\begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

all-zero rows at
the bottom

$\blacksquare$ = nonzero, $*$ = anything

Question

Is the identity matrix in echelon form?

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Answer: Yes

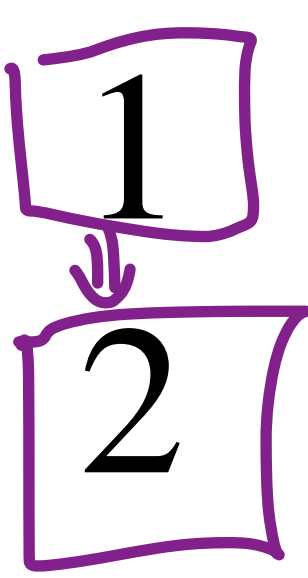
A 3x3 matrix is shown in row echelon form. The leading entries (1s) are highlighted with red boxes, and red arrows indicate the staircase pattern from the top-left to the bottom-right.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

the leading entries of each row appears to the right of the leading entry above it

it has no all-zero rows

Question

$$\begin{bmatrix} 2 & 3 & -8 \\ 0 & 1 & 2 \\ 0 & 2 & 0 \end{bmatrix}$$


Is this matrix in echelon form?

Answer: No

$$\begin{bmatrix} 2 & 3 & -8 \\ 0 & 1 & 2 \\ 0 & 2 & 0 \end{bmatrix}$$

The leading entry of the least row is not to the right of the leading entry of the second row

What's special about Echelon forms?

Thm. Let A be the augmented matrix of an *inconsistent* linear system. If $A \sim B$ and B is in echelon form then B has the row

$$[0 \ 0 \ \dots \ 0 \ 0 \ \blacksquare]$$

What's special about Echelon forms?

Thm. Let A be the augmented matrix of an *inconsistent* linear system. If $A \sim B$ and B is in echelon form then B has the row

$$[0 \ 0 \ \dots \ 0 \ 0 \ \blacksquare]$$

$$0 = 5 \quad \times$$

If all we care about is consistency then we just need to find an echelon form

Example

$$\begin{aligned}x - 2z &= 4 \\ -x + y + 5z &= -3 \\ x + 2y + 4z &= 7\end{aligned}$$

$$\begin{bmatrix} 1 & 0 & -2 & 4 \\ -1 & 1 & 5 & -3 \\ 1 & 2 & 4 & 7 \end{bmatrix} \quad R_2 \leftarrow R_2 + R_1$$

$$\begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 1 & 3 & 1 \\ -1 & 2 & 4 & 7 \end{bmatrix} \quad R_3 \leftarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 1 & 3 & 1 \\ 0 & 2 & 6 & 3 \end{bmatrix} \quad R_3 \leftarrow R_3 - 2R_2$$

$$\begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The Problem with Echelon Forms

If our system *is* consistent, we can't "read off" a solution from an echelon form

Reduced Echelon Form

Row-Reduced Echelon Form (RREF)

Definition. A matrix is in *(row-)reduced echelon form* if

1. The leading entry of each row appears to the right of the leading entry above it
2. Every all-zeros row appears below any non-zero rows
3. The leading entries of non-zero rows are 1
4. the leading entries are the only non-zero entries of their columns

Reduced Echelon Form (Pictorially)

leading entries are 1

$$\begin{bmatrix} 0 & 1 & * & 0 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

other column entries are 0

Basic Example

$$\begin{aligned}x_1 + x_3 &= 2 \\x_2 &= 1\end{aligned}$$

Basic Example

$$x_1 + x_3 = 2$$

$$x_2 = 1$$

$$x_1 = 2 - x_3$$

$$x_2 = 1$$

x_3 is free

The Fundamental Theorem

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Thm. *every* matrix is row equivalent to a *unique* matrix in reduced echelon form

The Fundamental Theorem

Thm. *every* matrix is row equivalent to a *unique* matrix in reduced echelon form

The Point. we can "read off" the solutions of a system of linear equations from its RREF

$$\begin{bmatrix} \boxed{1} & 0 & 0 & 5 \\ 0 & \boxed{1} & 0 & 1 \\ 0 & 0 & \boxed{1} & 3 \end{bmatrix}$$

How-To: Solving a System of Linear Equations

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1. Write your system as an augmented matrix

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2. Find the RREF of that matrix

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3. Read off the solution from the RREF

How-To: Solving a System of Linear Equations

1. Write your system as an augmented matrix

2. Find the RREF of that matrix

3. Read off the solution from the RREF

Our next topic

What's special about RREF?

Every leading variable can be written in terms of only non-leading variables

$$\begin{bmatrix} 0 & 1 & * & 0 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x + z = 2$$

$$y = 1$$

$$x = 2 - z$$

$$y = 1$$

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Forward Elimination and Back Substitution (Revisited)

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The goal of forward-elimination is to reduce matrix to an echelon form

Forward Elimination and Back Substitution (Revisited)

The goal of forward-elimination is to reduce matrix to an echelon form

The goal of back-substitution is to reduce an echelon form matrix to its **reduced** echelon form

General-Form Solutions

Q: How do we characterize all solutions, particularly in the infinite solution case?

Basic and Free Variables

Basic and Free Variables

The diagram shows an augmented matrix for a system of linear equations. The matrix is represented as $\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$. The first column, containing the pivot elements 1 and 0, is enclosed in a green oval and labeled "basic" in green. The second column, containing the pivot elements 0 and 1, is also enclosed in a green oval. The third column, containing the non-pivot elements 1 and 0, is enclosed in a blue oval and labeled "free" in blue. Above the first column is the label x_1 in red, above the second column is x_2 in red, and above the third column is x_3 in red. The rightmost column, containing the constants 2 and 1, is not enclosed in any oval.

$$x_1 + x_3 = 2$$
$$x_2 = 1$$

$$x_1 = 2 - x_3$$

$$x_2 = 1$$

x_3 is free

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The diagram shows a matrix in reduced echelon form:

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Annotations:

- x_1 is basic (points to the first column, which contains the pivot 1 in the first row)
- x_2 is basic (points to the second column, which contains the pivot 1 in the second row)
- x_3 is free (points to the third column, which does not contain a pivot)

Solutions of Reduced Echelon Forms

the row i of a pivot position describes the value of x_i in a solution to the system, in terms of the free variables

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

How-To: General Form Solution

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

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1. For each pivot position (i,j) , isolate x_j in the equation in row i

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1. For each pivot position (i,j) , isolate x_j in the equation in row i

2. If x_i is not in a pivot column then write

x_i is free

Example

$$x_1 + 2x_2 - 2x_4 = 4$$

$$x_3 + 3x_4 = 5$$

$$x_1 = 4 - 2x_2 + 2x_4$$

x_2 is free

$$x_3 = 5 - 3x_4$$

x_4 is free

$$x_1 = 4 - 2 + 10 = 12$$

$$x_2 = 1$$

$$x_3 = 5 - 3(5) = -10$$

$$x_4 = 5$$

$$\begin{array}{cccc|c} x_1 & x_2 & x_3 & x_4 & \\ \hline 1 & 2 & 0 & -2 & 4 \\ 0 & 0 & 1 & 3 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{array}$$

The matrix is shown with columns labeled x_1, x_2, x_3, x_4 above. The first column is enclosed in a green oval, the second in an orange oval, the third in a green oval, and the fourth in an orange oval. The first row has a purple box around the 1, the second row has a purple box around the 1, and the third row has a red box around the 0. A blue line is drawn under the third row.

x_1, x_3 basic

x_2, x_4 free

Question

$$\begin{bmatrix} 1 & 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 2 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Circle the pivot positions, highlight the pivot rows.

Which variables are free? Which are basic?

Write down a solution in general form for this reduced echelon form matrix.

*Write down a **particular** solution given the general form.*

Answer

$$\begin{bmatrix} 1 & 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 2 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = 1 - 3x_4$$

x_2 is free

$$x_3 = 4 - 2x_4$$

x_4 is free

Summary

Echelon form matrices "represent solutions"

General form solutions can be used to describe the infinite solution sets